

**SOME PROBLEMS
ON
THREE PHASE INDUCTION MOTOR**

1. A 746 kW, 3 phase, 50 Hz, 16 pole induction motor has a rotor resistance of $(0.02 + j 0.15)$ ohm at standstill. Full load torque is obtained at 360 rpm. Calculate (i) the ratio of maximum to full load torque (ii) the speed at maximum torque and (iii) the rotor resistance to be added to get maximum starting torque.
2. A 4 pole, 50 Hz, 3-phase induction motor develops a maximum torque of 162.8 N-m at 1365 rpm. The resistance of the star connected rotor is 0.2 ohm/phase. Calculate the value of the resistance that must be inserted in series with each rotor phase to produce a starting torque equal to half of the maximum torque.
3. A 440 volt, 3-phase, 50 Hz, 4-pole, Y-connected induction motor has a full load speed of 125 rpm. The rotor has an impedance of $(0.4 + j4)$ ohm and rotor/stator turn ratio of 0.8. Calculate (i) full load torque (ii) rotor current and full load rotor Cu-loss (iii) power output if windage and friction losses amount to 500 Watt (iv) maximum torque and speed at which it occurs (v) starting current and (vi) starting torque.
4. The rotor resistance and standstill reactance of a three phase induction motor are respectively 0.015 ohm and 0.09 ohm per phase. At normal voltage, the full load slip is 3%. Estimate the percentage reduction in stator voltage to develop full load torque at half full-load speed. Also, calculate the power factor.
5. A 18.65 kW, 4-pole, 50 Hz, 3-phase induction motor has friction and windage losses of 2.5 percent of the output. The full-load slip is 4%. Calculate for full load (a) the rotor Cu loss (b) the rotor input (c) the shaft torque (d) the gross electromagnetic torque.
6. A 400 volt, 4-pole, 3-phase, 50 Hz induction motor has a rotor resistance and reactance per phase of 0.01 ohm and 0.1 ohm respectively. Determine (a) maximum torque in N-m and the corresponding slip (b) the full load slip and power output in watts, if maximum torque is twice the full load torque. The ratio of stator to rotor turns is 4.

7. The maximum torque of a 3-phase induction motor occurs at a slip of 12%. The motor has an equivalent secondary resistance of 0.08 ohm/phase. Calculate the equivalent load resistance R_L , the equivalent load voltage V_L and the current at this slip, if the gross power output is 9,000 watts.
8. Find the ratio of starting to full load current in a 10 kW (output), 400 Volt, 3-phase induction motor with star-delta starter, given that full load power factor is 0.85, the full load efficiency is 0.88 and the blocked rotor current at 400 V is 40 A. Ignore magnetizing current.
9. A squirrel cage type induction motor when started by means of a star-delta starter takes 180% of full load line current and develops 35% of full load torque at starting. Calculate the starting torque and current in terms of full load values, if an auto-transformer with 75% tapping were employed.
10. At standstill, the equivalent impedance of inner and outer cages of a double cage rotor are $0.4 + j 2$ ohm and $(2 + j 0.4)$ ohm respectively. Calculate the ratio of torques produced by the two cages (i) at standstill and (ii) at 5% slip.
11. In a double cage induction motor if the outer cage has an impedance at standstill of $(2 + 1.2j)$ ohm, determine the slip at which the two cages develop equal torques if the inner cage has an impedance of $(0.5 + 3.5j)$ ohm at standstill.